

Hyperledger Sawtooth Transaction Families: Settings

This article is a continuation of our articles in the series on Hyperledger Sawtooth. It is primarily aimed at open source enthusiasts with an interest in blockchain technologies and a working knowledge of the Hyperledger Sawtooth project.



The Hyperledger Sawtooth Settings transaction family provides a method for storing on-chain configuration settings. The settings are stored in state, and as a result of this transaction family, play a critical role in the operation of a validator. For example, the consensus module uses these settings; in the case of PoET, one cross-network setting is the target wait time (which must be the same across validators), and this setting is stored as `sawtooth.poet.target_wait_time`. Other parts of the system use these settings similarly; for example, the list of enabled transaction families is used by the transaction processing platform. In addition, pluggable components such as transaction family implementations can use the settings during their execution. This design supports two authorisation options:

- A single authorised key which can make changes.
- Multiple authorised keys. In the case of multiple keys, a percentage of votes signed by the keys is required to make a change.

States in settings

This section describes how settings are stored and addressed

using the Settings transaction family. The settings data consists of settings/value pairs. A setting is the name for the item of the configuration data. The value is the data in the form of a string. Settings are name spaced using dots. For example:

```
sawtooth.poet.target_wait_time
sawtooth.validator.max_transactions_per_block
```

The Settings transaction family also uses settings for its own configuration. For example:

```
sawtooth.settings.vote.authorized_keys
sawtooth.settings.vote.approval_threshold
sawtooth.settings.vote.proposals
```

Definition of setting entries

The following protocol buffers define setting entries:

```
// Setting Container for the resulting state
message Setting {
```